Public Goods

<u>Externality of Consumption & Production</u> (Case of 2)

If there are certain kinds of externalities, it was not difficult to eliminate the inefficiencies.

In the case of a consumption externality between two people, for example, all one had to do was to ensure that initial property rights were clearly specified. People could then trade the right to generate the externality in the normal way.

In the case of production externalities, the market itself provided profit signals to sort out the property rights in the most efficient way. In the case of common property, assigning property rights to someone would eliminate the inefficiency.

<u>Externality of Consumption & Production (Case of More than 2)</u>

If there are more than two economic agents involved things become much more difficult. Suppose, for example, that instead of the two roommates, we had **three roommates**—one smoker and two nonsmokers. Then the amount of smoke would be a negative externality for both of the nonsmokers. **Let's suppose that property rights are well defined—say the nonsmokers have the right to demand clean air.** Although they have the right to clean air, they also have the right to trade some of that clean air away in return for appropriate compensation.

But now there is a problem involved

- the nonsmokers have to agree among themselves how much smoke should be allowed and
- what the compensation should be.

Perhaps one of the nonsmokers is much more sensitive than the other, or one of them is much richer than the other. They may have very **different preferences and resources**, and yet they both have to reach some kind of agreement to allow for an efficient allocation of smoke.

<u>Instead of roommates, we can think of inhabitants of a whole country</u>.

How much pollution should be allowed in the country?

If you think that reaching an agreement is difficult with only three roommates, imagine what it is like with millions of people.

The smoke externality with three people is an example of a public good—a good that must be provided in the same amount to all the affected consumers. In this case the amount of smoke generated will be the same for all consumers—each person may value it differently, but they all have to face the same amount.

Many public goods are provided by the government. For example, **streets and sidewalks** are provided by local municipalities. There are a certain number and quality of streets in a town, and everyone has that number available to use.

National defense is another good example; there is one level of national defense provided for all the inhabitants of a country. Each citizen may value it differently—some may want more, some may want less—but they are all provided with the same amount.

Public goods are an example of a particular kind of consumption externality: everyone must consume the same amount of the good.

Free Riding

Free riding is similar, but not identical. To see this, let us construct a numerical example of the TV problem described above. Suppose that each person has a wealth of $500, that each person values the TV at $100, and that the cost of the TV is $150. Since the sum of the reservation prices exceeds the cost, it is Pareto efficient to buy the TV. Let us suppose that there is no way for one of the roommates to exclude the other one from watching the TV and that each roommate will decide independently whether or not to buy the TV. Consider the decision of one of the roommates, Player A. If he buys the TV, he gets benefits of $100 and pays a cost of $150, leaving him with net benefits of −50. However, if Player A buys the TV, Player B gets to watch it for free, which gives B a benefit of $100. The payoffs to the game are depicted in Table 36.1.

Free riding game matrix.

		Player B	
		Buy	Don't buy
Player A	Buy	–50, –50	–50, 100
	Don't buy	100, –50	0, 0

The dominant strategy equilibrium for this game is for neither player to buy the TV. If player A decides to buy the TV, then it is in player B's interest to free ride: to watch the TV but not contribute anything to paying for it. If player A decides not to buy, then it is in player B's interest not to buy the TV either. This is similar to the prisoners' dilemma, but not exactly the same. In the prisoners' dilemma, the strategy that maximizes the sum of the players' utilities is for each player to make the same choice. Here the strategy that maximizes the sum of the utilities is for just one of the players to buy the TV (and both players to watch it).

If Player A buys the TV and both players watch it, we can construct a Pareto improvement simply by having Player B make a "sidepayment" to Player A. For example, if Player B gives Player A $51, then both players will be made better off when Player A buys the TV. More generally, any payment between $50 and $100 will result in a Pareto improvement for this example.

In fact, this is probably what would happen in practice: each player would contribute some fraction of the cost of the TV.

This public goods problem is relatively easy to solve, but more difficult free riding problems can arise in the sharing of other household public goods. For example, what about cleaning the living room? Each person may prefer to see the living room clean and is willing to do his part. But each may also be tempted to free ride on the other—so that neither one ends up cleaning the room, with the usual untidy results.

The situation becomes even worse if there are more than just two people involved —since there are more people on whom to free ride! Letting the other guy do it may be optimal from an individual point of view, but it is Pareto inefficient from the viewpoint of society as a whole.

What Is the Free Rider Problem?

The free rider problem is the burden on a shared resource that is created by its use or overuse by people who aren't paying their fair share for it or aren't paying anything at all.

The free rider problem can occur in any community, large or small. In an urban area, a city council may debate whether and how to force suburban commuters to contribute to the upkeep of its roads and sidewalks or the protection of its police and fire services. A public radio or broadcast station devotes airtime to fundraising in hopes of coaxing donations from listeners who aren't contributing.

KEY TAKEAWAYS

- Free riding is considered a failure of the conventional free market system.
- The problem occurs when some members of a community fail to contribute their fair share to the costs of a shared resource.
- Their failure to contribute makes the resource economically infeasible to produce.

Understanding the Free Rider Problem

The free rider problem is an issue in economics. It is considered **an example of a market failure.** That is, it is an inefficient distribution of goods or services that occurs when some individuals are allowed to consume more than their fair share of the shared resource or pay less than their fair share of the costs.

Free riding prevents the production and consumption of goods and services through conventional free-market methods. To the free rider, there is little incentive to contribute to a collective resource since they can enjoy its benefits even if they don't. As a consequence, the producer of the resource cannot be sufficiently compensated. The shared resource must be subsidized in some other way, or it will not be created.

When the Free Rider Problem Arises

The free rider problem as an economics issue only occurs under certain conditions:

- When everyone can consume a resource in unlimited amounts.
- When no one can limit anyone else's consumption.
- When someone has to produce and maintain the resource. That is, it's not a natural lake, it's a swimming pool, and someone had to undertake its construction and maintenance.

Economists point out that no business would voluntarily produce goods or services under these conditions. When the free rider problem looms, businesses back away. Either the shared resource will not be provided, or a public agency must provide it using taxpayer funds.

As an economic issue, the problem occurs when everyone can consume a resource in unlimited amounts, no one can limit anyone else's consumption, but someone has to produce and maintain the resource.

On the positive side, some people in every community will demonstrate that they feel a responsibility to pay their fair share. Some combination of a high sense of trust, positive reciprocity, and a sense of collective duty makes them willing to pay their fair share.

Beyond Economics

The free rider problem can crop up when the resource is shared by all and free to all. Like air. If a community sets voluntary pollution standards that encourage all residents to cut back on carbon-based fuels, many will respond positively. But some will refuse to make any change in their habits. If enough follow the standards, the air quality will improve and all the residents will benefit equally, even the free riders.

Solutions to the Free Riding Problem

Communities that face a free riding problem may try any of several solutions.

- Government addresses the problem by collecting and distributing tax dollars to subsidize public services. Theoretically, taxes are proportionate to income, so fair cost-sharing can be achieved.
- Communities can turn their public resource into a private or club resource, charging dues to make sure everyone who uses it contributes to it.
- Communities can impose a small fee on everyone. This will limit over-consumption and, over time, may even spur altruistic behavior. That is, many people may like the idea of making a small contribution to a resource that they use.

Public Goods and the Free Rider Problem

Public goods commonly face a free rider problem due to the two characteristics of a public good:

1. **Non-rival:** Consumption of the good or service by one individual does not reduce the availability of the good to others.

2. **Non-excludable:** It is impossible to prevent other consumers from consuming the good or service.

Examples of public goods include:

- National defense
- Fresh air
- Lighthouses
- Street lighting

Public goods create a free rider problem because consumers are able to utilize public goods without paying for them.